

MODULE-2

Cyber Crime and Cyber Law

Classification of cyber crimes, Common cyber crimes- cyber crime targeting computers and mobiles, cyber crime against women and children, financial frauds, social engineering attacks, malware and ransomware attacks, zero day and zero click attacks, Cybercriminals modus-operandi , Reporting of cyber crimes, Remedial and mitigation measures, Legal perspective of cyber crime, IT Act 2000 and its amendments, Cyber crime and offences.

Introduction

- Internet has undeniably opened a new way of exploitation known as cybercrime involving the use of computers, the Internet, cyberspace and the worldwide web (WWW).
- Figure 1, based on a 2008 survey in Australia, shows the cybercrime trend.
- While the worldwide scenario on cybercrime looks bleak, the situation in India is not any better.
- Indian corporate and government sites have been attacked or defaced more than 780 times between February 2000 and December 2002.
- A total of 3,286 Indian websites were hacked in 5 months – between January and June 2009.

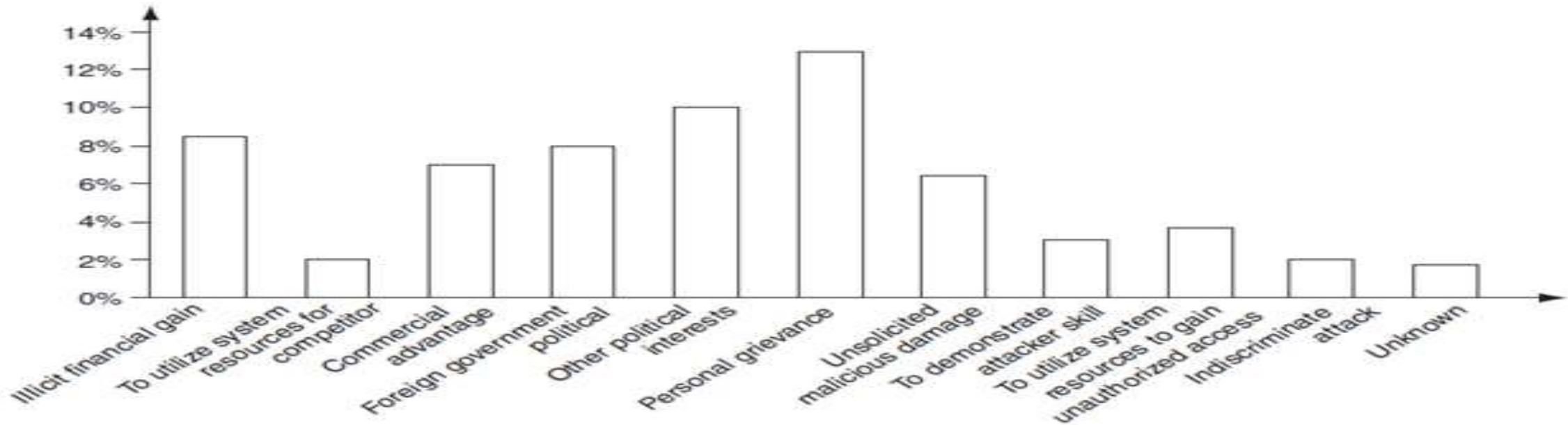


Figure 1

Cybercrime trend.

Source: 2008 Pacific Islands Computer Crime and Security Survey. Adapted from *Cybercrime: Threats, Challenges* presentation by Wipul Jayawickrama at the Computer Security Week 2008 in Brisbane, Australia (reproduced with permission).

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- **Cybercrime in India** has seen a significant rise, with a total of 65,893 cases registered in 2022, marking a 24.4% increase from the previous year. The majority of these cases were motivated by fraud, which accounted for 64.8% (42,710 cases) of the total cybercrimes.
- While detailed sector-wise data for all industries is not fully available, certain sectors have reported notable figures:
- **Education Sector:** This sector experienced an average of 2,507 cyberattacks per organization each week between 2022 and 2023.
- **Government and Military:** These sectors faced approximately 1,725 cyberattacks per week during the same period.
- **Healthcare Industry:** Healthcare organizations encountered around 1,684 weekly cyberattacks.
- **Retail and Wholesale:** This sector saw the highest year-over-year surge, with a 49% increase, averaging 1,079 attacks per week.

Cybercrime: Definition and Origins of the Word

The definitions of computer crime:

- 1. Any illegal act where a special knowledge of computer technology is essential for its perpetration, investigation or prosecution.
- 2. Any traditional crime that has acquired a new dimension or order of magnitude through the aid of a computer, and abuses that have come into being because of computers.
- 3. Any financial dishonesty that takes place in a computer environment.
- 4. Any threats to the computer itself, such as theft of hardware or software, sabotage and demands for ransom.
- The term “cybercrime” relates to a number of other terms such as:
 - *Computer-related crime*
 - *Computer crime*
 - *Internet crime*
 - *E-crime*
 - *High-tech crime*

Types of Cyber Attacks

Techno-crime: Also called **technology-enabled crime**, this refers to **criminal activities involving technology** as the main tool or target.

◆ Definition:

- Criminal acts that involve the use of technology—especially computers, the internet, or digital devices—to commit or facilitate illegal actions.
- A premeditated act against a system or systems, with the intent to copy, steal, prevent access, corrupt or otherwise deface or damage parts of or the complete computer system.

Examples:

- Hacking into systems to steal data
- Online financial fraud or scams
- Cyberstalking or harassment
- Identity theft using stolen credentials
- Piracy of software, movies, or games
- Child pornography or human trafficking via dark web

In India:

- Governed mainly by the **IT Act, 2000** and parts of the **IPC**.
- Increasing cases involve UPI frauds, fake online job offers, and phishing attacks.

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Techno-vandalism: This is a **subcategory of cybercrime** focused on **deliberate damage or disruption** to digital assets, websites, or systems.

Definition:

Intentional and unauthorized modification, defacement, or destruction of data, websites, or computer systems for mischief, protest, or malicious intent.

These acts of “brainless” defacement of websites and/or other activities, such as copying files and publicizing their contents publicly, are usually opportunistic in nature.

Examples:

- Defacing government or corporate websites (e.g., changing homepage to show protest messages)
- Disrupting a business’s operations by crashing its servers
- Launching denial-of-service (DoS) attacks to bring down platforms
- Injecting viruses to corrupt or erase data

Motives Can Be:

- Political (hacktivism)
- Personal (revenge, thrill)
- Criminal (diversion, ransom)

Differences between Techno and Techno-Vandalism crime

Aspect	Techno-Crime	Techno-Vandalism
Goal	Profit, control, or data theft	Disruption or defacement
Type	Broad (financial, identity theft, etc.)	Specific (digital destruction or mischief)
Examples	Phishing, ransomware	Website defacement, DDoS attacks
Damage	Economic, privacy, trust	Reputational, operational

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- Cybercrimes differ from most terrestrial crimes in four ways:
 - (a) how to commit them is easier to learn
 - (b) they require few resources relative to the potential damage caused
 - (c) they can be committed in a jurisdiction without being physically present in it
 - (d) they are often not clearly illegal.

Cyberterrorism is defined as “*any person, group or organization who, with **terrorist intent**, utilizes accesses or aids in accessing a computer or computer network or electronic system or electronic device by any available means, and thereby knowingly engages in or attempts to engage in a terrorist act commits the offence of cyberterrorism.*”

How cybercrimes are planned and how they actually take place

- Cyberterrorists usually use computer as a tool, target or both for their unlawful act to gain information.
- Internet is one of the means by which the offenders can gain priced sensitive information of companies, firms, individuals, banks and can lead to intellectual property (IP), selling illegal articles, pornography/child pornography, etc. This is done using:
 - Phishing, Spoofing, Pharming, Internet Phishing, wire transfer, etc.

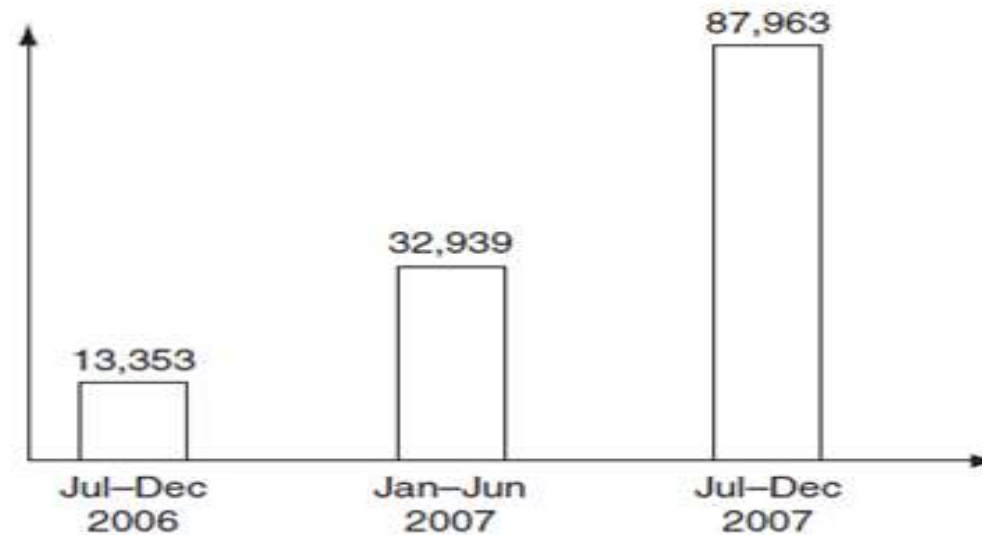


Figure 2 | Rise in the number of Phishing hosts.
Source: Symantec (International Telecommunications Society, 17th Biennial Conference, Montreal, Canada, June 24–27, 2008).

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- Figure 3 shows several categories of incidences – viruses, insider abuse, laptop theft and unauthorized access to systems.
- Typical network misuses are for:
 - Internet radio
 - streaming audio
 - streaming video
 - file sharing
 - instant messaging
 - Online gaming
 - Online gambling

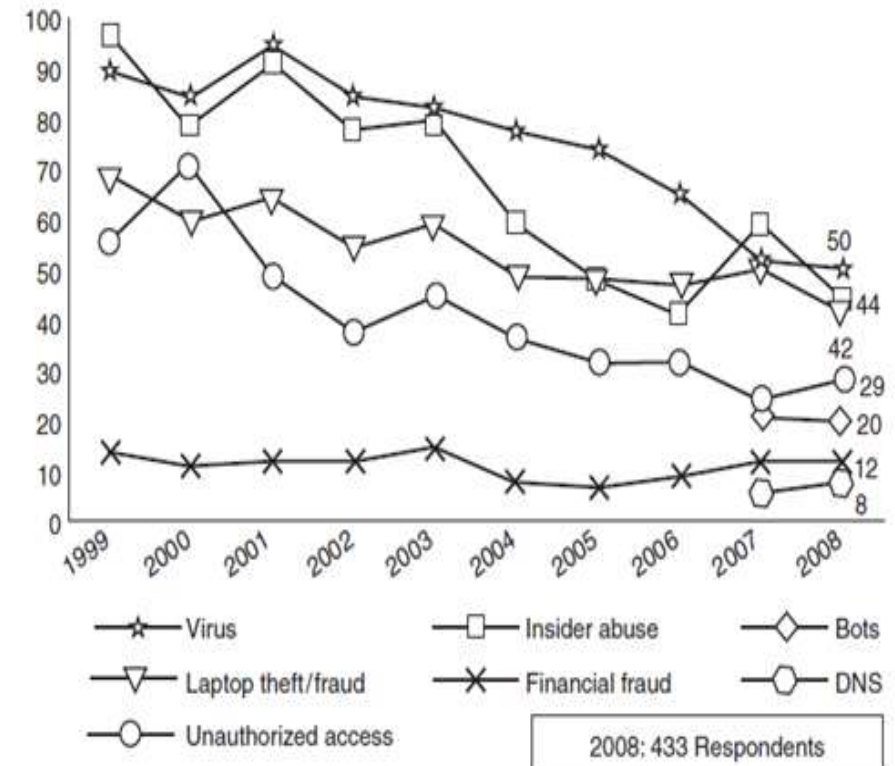


Figure 3 Major types of incidents by percentage.

Source: 2008 CSI Computer Crime and Security Survey available at the link <http://i.cmpnet.com/v2.gocsi.com/pdf/CSIsurvey2008.pdf> (15 March 2009).

What are cyber crimes

- Cyber crimes are crimes that involve criminal activities done through cyberspace by devices connected to the internet.
- At times, cyber crimes are also called 'computer crimes'.
- The major objective of committing such crimes is to gather confidential data from people and use it for monetary, political, or personal motives

Classification of Cyber Crimes

Classification of Cyber Crimes

Against individual

E Mail

Spamming

Cyber
Defamation

Harassment
Cyber Stalking

Against Property

Credit Card
Fraud

Intellectual
Property Crimes

Internet Time
Theft

Against Organisation

Unauthorised access

Denial of Service

Virus attack

E mail bombing

Salami attack

Logic bomb

Trojan horse

Data Didling

Against Society

Forgery

Cyber
terrorism

Web jacking

bercrimes

ad and narrow

Cybercrime in Broad Sense

Computer as a tool
The computer/information stored on the computer constitutes an important tool for committing the crime
Computer fraud, forgery, distribution of child pornography

Computer as the environment or context
The computer/information stored on the computer plays a non-substantial role in the act of crime, but does contain evidence of the crime
Murder using computer techniques, bank robbery and drugs trade pornography

ifications of Cybercrimes

Classifying cybercrimes – broad and narrow

	<i>Cybercrime in Narrow Sense</i>	<i>Cybercrime in Broad Sense</i>
Role of computer	<i>Computer as an object</i> The computer/information stored on the computer is the subject/target of the crime	<i>Computer as a tool</i> The computer/or information stored on the computer constitutes an important tool for committing the crime <i>Computer as the environment or context</i> The computer/information stored on the computer plays a non-substantial role in the act of crime, but does contain evidence of the crime
Examples	Hacking, computer sabotage, DDoS-attacks (distributed denial-of-service attacks), virtual child pornography	Computer fraud, forgery, distribution of child pornography Murder using computer techniques, bank robbery and drugs trade

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Cybercrimes are classified as follows:

- 1. Cybercrime against individual**
- 2. Cybercrime against property**
- 3. Cybercrime against organization**
- 4. Cybercrime against Society**
- 5. Crimes emanating from Usenet newsgroup**

Cybercrime against individual

- Electronic mail (E-mail) Spoofing and other online frauds
- Phishing: Spear Phishing and its various other forms such as Vishing and Smishing.
- Spamming
- Cyberdefamation
- Cyberstalking and harassment
- Computer sabotage
- Pornographic offenses
- Password sniffing: this also belongs to the category of cybercrimes against organization because the use of password could be by an individual for his/her personal work or the work he/she is doing using a computer that belongs to an organization.

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- Email spoofing: A spoofed email is one in which the e-mail header is forged so that the mail appears to originate from one source but actually has been sent from another source.
- Spamming: Spamming means sending multiple copies of unsolicited mails or mass e-mails such as chain letters.
- Cyber Defamation: This occurs when defamation takes place with the help of computers and/or the Internet. E.g. someone publishes defamatory matter about someone on a website or sends e-mails containing defamatory information.
- Harassment & Cyber stalking: Cyber Stalking Means following an individual's activity over internet. It can be done with the help of many protocols available such as e- mail, chat rooms, user net groups.

Cybercrime against property

- Credit card frauds
- Intellectual property(IP) crimes: Basically, IP crimes include software piracy; copyright infringement, trademarks violations, theft of computer source code, etc.
- Internet time theft

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- Credit Card Fraud: As the name suggests, this is a fraud that happens by the use of a credit card. This generally happens if someone gets to know the card number or the card gets stolen.
- Intellectual Property crimes: These include Software piracy: Illegal copying of programs, distribution of copies of software. Copyright infringement: Using copyrighted material without proper permission. Trademarks violations: Using trademarks and associated rights without permission of the actual holder. Theft of computer source code: Stealing, destroying or misusing the source code of a computer.
- Internet time theft: This happens by the usage of the Internet hours by an unauthorized person which is actually paid by another person

Cybercrime against organization

- Unauthorized accessing of computer: Hacking is one method of this and it is a punishable offense
- Password sniffing.
- Denial-of-service attacks (known as DoS attacks)
- Virus attack/dissemination of viruses
- E-Mail bombing/ mail bombs
- Salami attack/ Salami technique
- Logic bomb: Computer Sabotage
- Trojan Horse
- Data diddling
- Crimes emanating from Usenet newsgroup.
- Industrial spying/industrial espionage
- Computer network intrusions
- Software piracy

Contd..

- Unauthorized Accessing of Computer: Accessing the computer/network without permission from the owner. It can be of 2 forms: a) Changing/deleting data: Unauthorized changing of data. b) Computer voyeur: The criminal reads or copies confidential or proprietary information, but the data is neither deleted nor changed.
- Denial Of Service : When Internet server is flooded with continuous bogus requests so as to denying legitimate users to use the server or to crash the server.
- Computer contamination / Virus attack: A computer virus is a computer program that can infect other computer programs by modifying them in such a way as to include a (possibly evolved) copy of it. Viruses can be file infecting or affecting boot sector of the computer. Worms, unlike viruses do not need the host to attach themselves to.

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- Email Bombing: Sending large numbers of mails to the individual or company or mail servers thereby ultimately resulting into crashing.
- Salami Attack: When negligible amounts are removed & accumulated in to something larger. These attacks are used for the commission of financial crimes.
- Logic Bomb: It is an event dependent program. As soon as the designated event occurs, it crashes the computer, release a virus or any other harmful possibilities.
- Trojan Horse: This is an unauthorized program which functions from inside what seems to be an authorized program, thereby concealing what it is actually doing.
- Data diddling: This kind of an attack involves altering raw data just before it is processed by a computer and then changing it back after the processing is completed.

Cybercrime against Society

- Forgery
- Cyberterrorism
- Web jacking
- Crimes emanating from Usenet newsgroup: Usenet groups may carry very offensive, harmful, inaccurate or otherwise inappropriate material, or in some cases, postings that have been mislabeled or are deceptive in another way.

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- Forgery : Currency notes, revenue stamps, mark sheets etc. can be forged using computers and high quality scanners and printers.
- Cyber Terrorism : Use of computer resources to intimidate or coerce people and carry out the activities of terrorism.
- Web Jacking : Hackers gain access and control over the website of another, even they change the content of website for fulfilling political objective or for money.

Cyber Crimes Against Government

These crimes threaten national security and state institutions.

- **Hacking Government Websites** – Defacing or disrupting official portals.
- **Cyber Espionage** – Stealing classified information from government agencies.
- **Election Interference** – Manipulating online platforms to influence elections.

Impact of Cyber Crime

- **Financial Loss** – Billions of dollars are lost annually due to fraud, ransomware, and online scams.
- **Data Breaches** – Sensitive personal and corporate information can be stolen and misused.
- **Reputation Damage** – Businesses and individuals can suffer reputational harm due to leaked information.
- **Legal Consequences** – Governments worldwide have enacted strict laws against cyber crimes, with severe penalties for offenders.

Common cyber crimes

1. Hacking

Hacking refers to gaining **unauthorized access** to a computer system, network, or data.

- It can be done by exploiting system vulnerabilities.
- Hackers may **steal, modify, or delete** sensitive data.
- Types of hacking:
 - **Ethical Hacking** – Done for security testing (legal).
 - **Black Hat Hacking** – Malicious activities for personal gain.
 - **White Hat Hacking** – Used to improve security legally.



Example:

- A hacker breaks into a corporate server and steals customer information for financial fraud.

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2. Phishing

Phishing is a type of cyber crime where attackers send fraudulent **emails, messages, or websites** to trick users into revealing personal information.

- Usually targets **passwords, banking details, or credit card numbers**.
- Often disguised as legitimate requests from banks, social media, or organizations.

Example:

- A fake email from a "bank" asking users to update their account credentials, leading to identity theft



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3. Identity Theft

Identity theft occurs when someone **steals and misuses another person's identity** to commit fraud.

- Cyber criminals use stolen identities for **bank fraud, online purchases, and loan applications.**
- Often done using phishing attacks, hacking, or social engineering.



Example:

A criminal uses stolen personal information **to open a credit card account** under a victim's name.

4. Cyber Stalking

Cyber stalking involves **harassing, threatening, or intimidating** someone through digital platforms.

- Common methods include **emails, social media, or tracking software.**
- It is a serious offense and can lead to **psychological trauma** for victims.

Example:

A person continuously **sends unwanted messages and threats to someone** on social media.



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5. Cyber Bullying

Cyber bullying refers to the **use of electronic communication** to harass, embarrass, or threaten someone.

- Occurs mostly among **teenagers and young adults**.
- Includes **spreading false information, posting humiliating content, or sending threats**.

Example:

- A student is bullied on social media **when someone posts edited images to humiliate them**



6. Ransomware Attacks

Ransomware is a **malicious software** that encrypts files and demands **ransom payments** to restore access.

- Victims include **individuals, businesses, hospitals, and government organizations**.
- Attackers demand payment via **cryptocurrency** to remain anonymous.

Example:

- A company's files **are encrypted, and hackers demand a ransom**
- to unlock them.



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7. Online Scams and Fraud

Cyber criminals use fake websites, emails, or messages to **trick victims into financial loss**.

- Common scams include:
 - **Lottery & Prize Scams** – Claiming the victim won money.
 - **Fake Job Offers** – Asking for payments before providing a job.
 - **Investment Fraud** – Promising high returns on fake schemes.



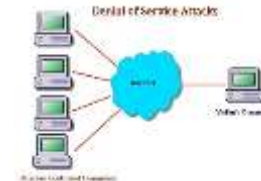
Example:

A person **receives an email saying they won a lottery and must pay a fee to claim** the prize.

8. Distributed Denial of Service (DDoS) Attacks

DDoS attacks flood a website or server with **excessive traffic**, making it unavailable.

- Used to disrupt businesses, governments, and financial services.
- Often executed by a **botnet** (a network of infected computers).



Example:

A cybercriminal **uses a botnet to crash an e-commerce website** during a major sale.

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9. Child Exploitation and Cyber Pornography

Cyber criminals use the internet to **create, distribute, or access illegal content involving minors.**

- Includes **child pornography, grooming, and abuse.**
- International laws and cyber police track such activities.



Example:

A criminal shares illegal content on the dark web.

10. Cyber Terrorism

Cyber terrorism involves **attacking government infrastructure, financial systems, or national security.**

- Targets include **power grids, banking systems, and military networks.**
- Attackers can be **hacktivists, enemy nations, or criminal organizations.**



Example:

A group of hackers attacks a country's financial system to disrupt its economy.



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11. Software Piracy

Software piracy is the **unauthorized copying, distribution, or use of copyrighted software**.

- Reduces profits for developers and companies.
- Includes **counterfeit software, cracks, and illegal downloads**.



Example:

A user **downloads and installs a cracked version of an expensive** software without a license.

12. Espionage and Corporate Data Theft

Cyber criminals or rival organizations **steal confidential business data** to gain an advantage.

- Includes **trade secrets, financial records, and customer databases**.

Example:

A company's employee sells confidential project details to a competitor.



Cyber crime targeting computers and mobiles

- Cybercrime targeting computers and mobile devices is a growing concern in today's digital world.
- These crimes encompass a wide range of illegal activities conducted using technology, often with the goal of financial gain, data theft, or causing harm to individuals, organizations, or governments.
- The computer may have been used in the execution of a crime or it may be the target.
- Cybercrime can be defined as “The illegal usage of any communication device to commit or facilitate in committing any illegal act”.
- A cybercriminal is a person who uses his skills in technology to do malicious acts and illegal activities known as cybercrimes. They can be individuals or teams.

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- Most cybercrime falls under two main categories:
 - Criminal activity that *targets computers*.
 - Criminal activity that *uses computers*.
- Cybercrime that *targets* computers often involves malware like viruses.
- Cybercrime that *uses* computers to commit other crimes may involve using computers to spread malware, illegal information or illegal images.
- Cyber crimes targeting computers and mobile devices involve attacks that exploit vulnerabilities in software, hardware, or networks to steal data, disrupt services, or gain unauthorized access.

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1. Cyber Crimes Targeting Computers

A. Hacking & Unauthorized Access

- Gaining **unauthorized entry** into a computer system or network.
- Attackers may **steal, alter, or delete data** for financial or personal gain.
- Methods include:
 - **Brute force attacks** – Guessing passwords using automated scripts.
 - **SQL injection** – Injecting malicious queries to manipulate databases.
- **◆ Example:** A hacker accesses a corporate database and steals customer credit card details.

How to Prevent Unauthorized Access & Hacking

Defense Type	What You Can Do
 Strong Passwords	Use long, unique passwords and change them regularly.
 Multi-Factor Auth	Require MFA for all users – even if a password is stolen, it can't be used alone.
 Regular Updates	Patch software and systems quickly to fix known vulnerabilities.
 Monitor & Detect	Use tools like firewalls, intrusion detection systems (IDS), and activity logs to spot unusual behavior.
 Security Training	Teach employees how to spot phishing and avoid bad links or downloads.
 Backup Data	Regularly back up critical systems offline so recovery is possible without paying ransom.
 Access Control	Give users access only to the data and systems they need (principle of least privilege).

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B. Malware Attacks (Viruses, Worms, Trojans, Ransomware)

Malware is malicious software designed to **damage or control** a system.

- **Viruses** – Attach to files and spread when executed.
 - **Worms** – Spread across networks without human intervention.
 - **Trojans** – Disguised as legitimate software but allow unauthorized access.
 - **Ransomware** – Encrypts files and demands ransom for decryption.
- ◆ **Example:** A ransomware attack locks a hospital's patient records, demanding payment to restore access.

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Example: WannaCry Ransomware Attack (2017) :Over 200,000 computers in 150+ countries

Major included was

- **UK's National Health Service (NHS)**
- **FedEx**
- **Renault**
- **Telefonica (Spain)**

What Happened:

WannaCry is a ransomware worm that spread across computers using a vulnerability in Windows systems called **EternalBlue** (originally developed by the NSA, later leaked).

Once inside, WannaCry encrypted files and displayed a message demanding **\$300 in Bitcoin**, doubling after a few days if unpaid.

It also spread automatically to other computers on the same network, causing mass infection in hours.

How it worked:

Exploited a Windows vulnerability in SMB protocol.

Installed ransomware payload and encrypted files.

Displayed ransom note to the victim.

Spread to other systems using worm-like behavior.

Impact (Example: NHS UK)

Hospitals had to cancel surgeries and appointments.

Ambulances were diverted.

Patient records became inaccessible.

Estimated cost: Over \$100 million in damages to NHS alone.

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C. Distributed Denial of Service (DDoS) Attacks

- Overloading a **website or server with excessive traffic**, making it inaccessible.
- Often executed using a **botnet** (network of infected devices).
- **Example:** A DDoS attack disrupts an online banking website, preventing transactions.

D. Cyber Espionage & Data Theft

- **Spying on organizations or governments** to steal sensitive data.
- Often conducted by **competitors, criminals, or state-sponsored hackers**.
- **Example:** A competitor hacks a company's servers to steal trade secrets.

E. Cyber Vandalism (Website Defacement & Destruction)

- Hackers modify or deface **websites and digital content**.
- May be done for **political, ideological, or personal motives**.
- **Example:** A hacktivist group replaces a government website with protest messages.

2. Cyber Crimes Targeting Mobile Devices

A. Mobile Malware & Spyware

- Malware designed for **Android and iOS devices**.
- Often spreads through **fake apps, infected links, or email attachments**.
- **Spyware** records **keystrokes, messages, and GPS location**.
- **◆ Example:** A fake antivirus app secretly records the user's banking passwords.

B. SIM Card Cloning & Mobile Hacking

- Attackers clone a **SIM card to intercept messages and calls**.
- Can be used for **financial fraud and unauthorized access**.
- **◆ Example:** A hacker clones a victim's SIM to receive OTPs and access bank accounts.

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C. Bluetooth & Wi-Fi Hacking

- Attackers exploit **unsecured Bluetooth and Wi-Fi networks** to gain control.
- **Man-in-the-Middle (MITM) Attacks** allow hackers to intercept data.
- ♣ **Example:** A hacker intercepts login credentials when a user connects to a free public Wi-Fi.

D. Smishing (SMS Phishing)

- Cyber criminals send fraudulent **SMS messages** to trick users into clicking malicious links.
- Often used to **steal passwords, banking details, or install malware**.
- ♣ **Example:** A victim receives an SMS pretending to be from their bank, asking for account details.

E. Mobile Ransomware & Lock Screen Attacks

- Locks the mobile screen and demands **ransom to regain access**.
- Often disguised as a system update or antivirus app.
- ♣ **Example:** A ransomware app locks a phone, displaying a fake police message demanding payment.

Web Jacking

- Web Jacking occurs when someone forcefully takes control of a website (by cracking the password and later changing it).
- Thus, the first stage of this crime involves “password sniffing.”
- The actual owner of the website does not have any more control over what appears on that website.

Forgery

- Counterfeit currency notes, postage and revenue stamps, marksheets, etc. can be forged using sophisticated computers, printers and scanners.
- These are made using computers and high quality scanners and printers.

Denial-of-Service (DoS) and Distributed Denial-of-Service (DDoS) Attacks

- The goal of the Denial-of-Service attack is to make the service unavailable by flooding or crashing the system with voluminous traffic that the server cannot accommodate. In DoS, a single hacker/attacker penetrates the victim's system; whereas, in DDoS, multiple attackers penetrate the victim's system.

List of DoS Attacks

- **Buffer overflow attack:** A buffer is a physical area of storage that holds temporary data when moved from one place to another. Buffer can hold only a defined amount of data. When it exceeds the limit, it overwrites the memory adjacent to it causing the program to misbehave. An attacker uses this as an advantage to deliberately overwrite or replace the code in the buffer, which eventually causes the whole program to misbehave so that the attacker can take control of it.
- **ICMP flood/Ping flood:** Here, the attacker floods the target machine with ICMP 'echo requests' known as the pings. This results in Denial-of-Service to the target server.

- **SYN flood:** It is a protocol attack. SYN (synchronize) is the first step when you establish communication between two systems over the TCP/IP protocol.
- The receiver responds with an ACK (acknowledge) message on the successful connection.
- In SYN flood, the attacker initiates the connection and sends a massive number of SYN requests, but it does not get any ACK message back.
- This makes the server spend resources on the waiting responses till enough resources are consumed.
- The result is that the server becomes unresponsive to handle the legitimate traffic.

Teardrop attack: In this type, the attacker sends segmented packets to a target machine.

- The TCP/IP protocol cannot reassemble fragmented packets, which results in an overlap of packets and eventually causes the target network machine to crash.

- **Smurf attack:** Here, the attacker floods the target server with Internet Control Message Protocol (ICMP) packets.
- The request is made from the spoofed IP of the target device to multiple network devices.
- When the devices respond, it amplifies the initial ICMP flood attack and overwhelms the target, making it unresponsive.
- **HTTP flood:** In HTTP flood, the attacker exploits the HTTP GET or POST requests to bombard the **web_server**/application.
- **Ping of death:** IN this type, the attacker pings packets, which are larger than the maximum size, to disrupt the target machine and to crash it or freeze it.

Botnet: A botnet refers to a group of computers which have been infected by malware and have come under the control of a malicious actor.

- A single attacker called the 'bot herder' takes control of a network of computers and exploits it.
- Here, the attacker acts as a main lead who commands every bot to carry out illegal actions coordinately.
- A bot refers to the single target machine and botnet refers to a group of bots which are under the command of bot herder who can control millions of bot at a time.
- When the attacker instructs for any updates to cause misbehaviour, all the bots receive simultaneously.

Drive-by Attack

- When the system has security flaws due to a lack of updates on OS, app, or browser, an attacker can trigger the unintentional download of malicious code to the targeted computer or mobile device, making it vulnerable.
- In this attack, the victim does not necessarily have to click on any links, open a malicious email attachment, or download any files.

Prevention Strategies

✓ For Computers:

- Install **antivirus and firewalls**.
- Keep **software updated**.
- Avoid **clicking on unknown links or attachments**.
- Use **strong passwords and two-factor authentication**.

✓ For Mobile Devices:

- Download apps only from **official stores**.
- Disable **Bluetooth and Wi-Fi** when not in use.
- Be cautious with **SMS messages and unknown numbers**.
- Regularly check **app permissions** and remove unnecessary apps.

Cyber crime against women and children

Cybercrimes against women and children are particularly concerning because they often involve harassment, exploitation, or abuse of vulnerable individuals.

Cybercrime against women and children in India, including Karnataka, has seen a significant rise in recent years. These crimes encompass various forms, such as online harassment, cyberstalking, and the distribution of explicit content.^[2]

National Overview:

Increase in Cybercrime Cases: According to the National Crime Records Bureau (NCRB), cybercrime cases in India surged from 52,974 in 2021 to 65,893 in 2022. Notably, complaints related to cybercrimes against women also saw an uptick during this period. ^[cite?turn0search19?]

Crimes Against Children: A decade-long analysis reveals a 179% increase in crimes against children between 2013 and 2022, with over 445 incidents occurring daily across the country

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Karnataka Specifics:

- **Digital Arrest Scams:** In 2024, Karnataka reported 641 cases of digital arrest scams, leading to losses amounting to ₹109 crore. Bengaluru was the most affected city, followed by Mysuru and Mangaluru. These scams often targeted senior citizens and women, with fraudsters impersonating officials. ⁷

Cybercrime Statistics: Mysuru climbed to the fourth position in cybercrime cases within Karnataka for 2024, with a notable number of cases still under investigation.

Government Initiatives:

National Cyber Crime Reporting Portal: The Government of India has launched the National Cyber Crime Reporting Portal (<https://cybercrime.gov.in>) under the Cyber Crime Prevention against Women and Children (CCPWC) scheme. This platform enables the public to report incidents related to all types of cybercrimes, with a special focus on those against women and children. As of April 2024, the portal received over 1.94 lakh complaints related to Child Pornography and Rape or Gang Rape content.

Contd..

Case Highlight:

International Collaboration: In July 2024, an Indian man was arrested for allegedly sexually abusing and intimidating a girl from Western Australia over Instagram and WhatsApp. The abuse began in 2020 when the girl was 11 years old. This case underscores the global nature of cybercrimes and the importance of international cooperation in addressing them.

Women are most commonly exposed to the following Cyber Crimes:

- **Sextortion:** The most common cybercrime performed against women during the pandemic was sextortion.
- By using their victims' private photos or altered images as blackmail, the offenders started demanding money or sexual favors from them.
- In order to express their aggravation about the epidemic, the offenders threatened women and asked for sexual videoconferencing or letters from them.
- Additionally, as they had no money, they felt empowered to threaten victims with their altered images in order to get money from them.

Contd..

- **Pornography:** This criminal activity involves posting morphed images of victims and using them for pornographic purposes, sometimes demanding money to remove them from social networking sites.
- Pornographic Offenses
- “Child pornography” means any visual depiction, including but not limited to the following:
 1. Any photograph that can be considered obscene and/or unsuitable for the age of child viewer;
 2. film, video, picture;
 3. computer-generated image or picture of sexually explicit conduct where the production of such visual depiction involves the use of a minor engaging in sexually explicit conduct.
- Child pornography is considered an offense.
- Unfortunately, child pornography is a reality of the Internet.
- The Internet is being highly used by its abusers to reach and abuse children sexually, worldwide.
- In India too, the Internet has become a household commodity in the urban areas of the nation.

Contd..

- **Cyber Grooming:** In this case, a person builds a relationship with a woman through an online platform and pressurizes her for undue favors or doing sexual acts.
- **Cyber stalking:** It included, among other things, contacting or trying to engage the victim via social media sites or phone conversations despite her obvious lack of interest, posting messages on the victim's page (often threatening in nature), and persistently bothering the victim with emails and phone calls to create mental distress.

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Cyber hacking: During the pandemic, people started reading the news online.

- There are more examples of false news and information now than ever before.
- After clicking on malicious URLs, the women were the victims of cyber hacking.
- The malware downloaded all of their personal information to their phones, turned on the microphone and camera, and took their intimate photos and videos.
- Then, criminals use these details carry out extortion and other offenses.

Contd..

- **Cybersex trafficking:** It is different from physical sex trafficking in that the victim does not physically engage with the perpetrator.
- Cybersex trafficking is when a dealer broadcasts, records, or takes pictures of the victim engaging in sexual or intimate activities from a central location and then sells the content online to sexual predators and clients.
- The criminals have forced, manipulated, and blackmailed women into participating in cybersex trafficking, which constitutes sexual abuse of women.

Financial frauds

- Financial frauds can have devastating consequences for individuals and the economy as a whole. While digital payments have made life convenient and easy in India, they have also made us prone to all kinds of financial frauds.

Ponzi Schemes: A Mirage of False Promises

Ponzi schemes lure investors with promises of unusually high returns in a short period. The fraudsters use funds from new investors to pay off earlier investors, creating a false illusion of profitability.

One infamous example is the Saradha chit fund scam, where millions of investors lost their hard-earned money.

The group, consisting of over 200 private companies, falsely portrayed its collective investment schemes as chit funds.

With an estimated collection of ₹200 to 300 billion (US\$4–6 billion), the scheme managed to attract deposits from more than 1.7 million individuals before its eventual downfall.

Identity fraud

Identity fraud is common on Internet. Criminals have a few options when it comes to stealing your sensitive information.

How do you know you're being targeted?

- Unfamiliar transactions on your credit card.

- Strange charges on your bank statements.

- New credit cards or loans in your name.

- Missing or error-filled tax returns.

- Calls from debt collectors about purchases you didn't make.

- A drop in credit score.

- Bounced checks.

Contd..

Fraudulent charities

Scammers use philanthropy as fraud, too. Charity fraud entails creating a fake charity and collecting “donations” that disappear along with the thief.

How does charity fraud happen?

Scammers create fake charities like military veteran charities that sound like ones you know and trust. These scams are especially common during natural disasters or international news events.

Credit card fraud

There are several ways that criminals can steal your credit card information. They could steal your physical card, trick you into entering information on a phishing website or email, buy your details on the Dark Web, or use any number of other credit card scams.

Hackers can also create a clone of your physical card using just your credit card numbers.

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Stock Market Manipulation

Stock market manipulation includes activities like price rigging, spreading false information, insider trading, and pump-and-dump schemes. Fraudsters manipulate stock prices, deceiving investors and causing significant financial losses.

The Satyam Computer Services scandal is a prime example, where the company's promoters manipulated financial statements to inflate stock prices.

Bank Frauds

Bank frauds encompass various fraudulent activities, including loan frauds, cheque frauds, forged documents, and unauthorized transactions. These frauds result in substantial financial losses for banks and individuals. • One notable case is the Nirav Modi-PNB scam, where fraudulent Letters of Undertaking were issued, causing a massive loss to Punjab National Bank

Social Engineering Attacks

Social Engineering

- It is the “technique to influence” & “persuasion to deceive” people to obtain the information.
- It exploits the fact that people are the weak link in security.
- Social engineers build the trust with the victim/person to gain the unauthorized information/access
- Their goal is to fool someone into providing valuable information.

Example: The attacker (social engineer) calling a user & pretending to be a tech support person & ask questions about the confidential files, passwords, etc.

Classification of Social Engineering

1. Human based Social Engineering:

- It refers to person to person interaction to get the unauthorized information.
- The following are its different types.
 - i. Impersonating an employee or valid user: The attacker impersonates/poses as an employee of the same organization to take the advantage from the people who are helpful.
 - ii. Posing as important user: The attacker pretends to be a CEO/Manager who intimidates lower level employee in order to gain access to the system.
 - iii. Using a third person: The attacker pretends to have permission from an authorized source/person (who cannot be contacted for verification) to use a system.
 - iv. Calling technical support: Attacker calls help desk or tech support personnel to obtain the information since they are trained to help users.
 - v. Shoulder surfing: It involves gathering information (usernames, passwords, etc) by watching over a person's shoulder while he/she logs into the system.
 - vi. Dumpster diving (Scavenging/Binning): It involves looking in the trash/dustbin for information written on pieces of paper, computer print outs, etc.

Contd..

2.Computer based Social Engineering

- It refers to the attempts made to get the unauthorized information by using computer/software/internet.
- The following are its different types.
 - i. Fake emails: It involves the attacker sending fake emails (pretending as a legitimate email) to a number of users in order to make the users to reveal their sensitive information such as usernames, passwords, credit card details, etc. It is also called as Phishing.
 - ii. Email attachments: It involves sending malicious codes to victim's system in the form of an email attachment. The virus, worms, etc which will be present in the email attachment will be automatically executed if the victim opens the attachment.
 - iii. Pop-up windows: They are used similar to email attachments but they encourage the victim to click on special offers or free stuffs so that the malicious code can be installed to the system

Effects of Social Engineering:

Loss/altering of medical & healthcare information, corporate financial data, electronic funds transfers, etc.

Loss of customers

Loss of funds

Loss of trust

Collapse of the organization

Malware and Ransomware attacks

Malware attacks: Malware attacks are any type of malicious software designed to cause harm or damage to a computer, server, client or computer network and/or infrastructure without end-user knowledge

Types of Malware:

1. Adware
2. Virus
3. Trojans
4. Worms
5. Bots
6. Downloaders
7. Remote Access Tool(RAT)
8. KeyLoggers

Ransomware

- A ransomware attack is a malware that encrypts personal information and documents while demanding a ransom amount to decrypt them. ▪ Once the files are encrypted or locked behind a password, a text file is available to the victim, explaining how to make the ransom payment and unlock the files for it.

How Does a Ransomware Attack Work?

- The spread of ransomware mostly starts with phishing attacks. A ransomware attack gains access to a victim's device through infected emails, messages, and malicious sites and encrypts the data in that device.
- The ransomware uses simple asymmetric encryption algorithms, blocks a user's files, and makes them difficult to decrypt without knowing the key.
- Another way to breach a system with ransomware is by using the Remote Desktop Protocol or RDP access. It can access remotely a computer using this protocol allowing a hacker to install malicious software on the system with the owner, unaware of these developments.
- Ransomware adds instruction files describing the pay-for-decryption process, then uses those files to present a ransom note to the user.
- Ransomware usually terminates and destroys itself by leaving only the payment instruction files.

Types of Ransomware

1. Locker ransomware

It is a type of malware that blocks standard computer functions from being accessed until the payment to the hackers is not complete.

It shows a lock screen that doesn't allow the victim to use the computer for primary purposes.

2. Crypto ransomware

This ransomware encrypts the local files and documents on the computers.

Once the files are encrypted, finding the decryption key is impossible unless the ransomware variant is old and the keys are already available on the internet.

3. Scareware

It is a fake software that claims to have detected a virus or other issue on your computer and directs you to pay to resolve the problem.

Some scareware locks the computer, while others flood the screen with popup alerts without damaging files.

How to Prevent Ransomware Attacks?

- One must always have backups of their data. Cloud storage for backup is easy, but a physical backup in a hard drive is always recommended.
- Keeping the system updated with the latest security patches is always a good idea.
- Apart from system updates, one must always have reputed antivirus software installed.
- If a system is infected with ransomware already, there is a website, 'nomoreransom.org.' It has a collection of decryption tools for most wellknown ransomware packages.

Zero day and Zero click attacks Zero day

- Software often has security vulnerabilities that hackers can exploit to cause havoc.
- The term "zero-day" refers to the fact that the vendor or developer has only just learned of the flaw – which means they have “zero days” to fix it.
- A zero-day attack takes place when hackers exploit the flaw before developers have a chance to address it.
- Zero-day attackers can steal data, corrupt files, take control of devices, install malware or spyware, and more.

Typical targets for a zero-day exploit include:

1. Government departments.
2. Large enterprises.
3. Individuals with access to valuable business data, such as intellectual property.
4. Hardware devices, firmware and Internet of Things (IoT)

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Recent Examples of Zero Day Attacks

- In December 2021, Amazon Web Services, Microsoft, Cisco, Google Cloud, and IBM were among the major tech players affected by the Log4j vulnerability in an opensource logging library.
- In 2021, Google's Chrome suffered a series of zero-day threats, causing Chrome to issue updates. The vulnerability stemmed from a bug in the V8 JavaScript engine used in the web browser.
- Zoom was targeted in 2020. Hackers were able to remotely access users' PCs if the video conferencing platform was running on an older version of Windows.
- Apple's iOS fell victim in 2020 to two sets of zero-day bugs that saw attackers compromising iPhones remotely.

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How to protect yourself against zero-day attacks

1. Keep all software and operating systems up to date. This is because the vendors include security patches to cover newly identified vulnerabilities in new releases. Keeping up to date ensures you are more secure.
2. Use only essential applications. The more software you have, the more potential vulnerabilities you have. You can reduce the risk to your network by using only the applications you need.
3. Use a firewall. A firewall plays an essential role in protecting your system against zero-day threats. You can ensure maximum protection by configuring it to allow only necessary transactions.

Zero click

- Zero-click attacks require no action from the victim – meaning that even the most advanced users can fall prey to serious cyber hacks and spyware tools.
- Also called interaction-less or fully remote attacks.
- Spying software relies on convincing the targeted person to click on a compromised link or file to install itself on their phone, tablet, or computer.
- As a result, zero-click malware or no-click malware is much more dangerous.
- The target of a zero-click attack can be anything from a smartphone to a desktop computer and even an IoT device.

Examples of Zero-Click Attacks

1. Apple zero-click, forced entry, 2021: In 2021, a Bahraini human rights activist had their iPhone hacked by powerful spyware sold to nation-states.
2. WhatsApp breach, 2019: This infamous breach was triggered by a missed call, which exploited a flaw in the source code framework of WhatsApp.

How to protect yourself from zero-click exploits

- Keep your operating system, firmware, and apps on all your devices up to date as prompted.
- Only download apps from official stores. ▪ Delete any apps you no longer use.
- Use your device password protection.
- Use strong authentication to access accounts, especially critical networks.
- Use strong passwords – i.e., long and unique passwords.

Modus Operandi of Cyber Criminals

- In general, modus operandi is the method acquired by any criminal for the successful commission of a crime.
- At a minimum, every Modus Operandi will contain three basic elements namely:
 1. Ensure success of the crime
 2. Protect identity
 3. Facilitate effective escape

Common forms of modus operandi:

- Sending Annoying Messages
- Making Offensive Calls

Reporting of cyber crimes

- Reporting cybercrimes is essential to combat online threats and hold perpetrators accountable.

Here are the steps you can take to report cybercrimes:

1. Contact Your Local Law Enforcement:
2. Report to a National Cybersecurity Agency
3. Report to the Appropriate Online Platforms:
4. Report to Anti-Fraud Organizations: There are organizations like the AntiPhishing Working.
5. Report to Financial Institutions:
6. Report to Internet Service Providers (ISPs)
7. Document the Incident:
8. Use Online Reporting Portals:
9. Consider Legal Advice:
10. Protect Yourself:

Remedial and mitigation measures:

Remedial Measures:

1. Incident Response: In the event of a cyber crime, organizations should have an incident response plan in place to quickly identify, contain, and mitigate the impact of the attack. This includes isolating affected systems, restoring backups, and applying patches or security updates.
2. Forensic Investigation: Engaging professional forensic investigators can help identify the source and extent of the cyber crime, gather evidence, and aid in legal proceedings.
3. Data Recovery: If data is compromised or encrypted due to a cyber attack, organizations should have backups in place to restore affected systems and minimize data loss.

- Mitigation Measures:

1. Strong Security Practices: Implement robust security measures, such as firewalls, antivirus software, and intrusion detection and prevention systems, to protect against cyber threats.
2. Regular Updates and Patching: Keep software, operating systems, and firmware up to date with the latest security patches to mitigate vulnerabilities that cyber criminals may exploit.
3. Employee Education: Provide cybersecurity awareness and training programs to employees to educate them about common cyber threats, phishing techniques, and safe online practices.
4. Multi-factor Authentication (MFA): Implement MFA wherever possible to add an extra layer of security, making it harder for cyber criminals to gain unauthorized access to accounts or systems.
5. Data Encryption: Encrypt sensitive data, both in transit and at rest, to ensure that even if it is intercepted or stolen, it remains unreadable and unusable for unauthorized individuals.
6. Regular Security Audits: Conduct regular security audits and vulnerability assessments to identify and address any weaknesses or potential entry points for cyber criminals.

Legal perspective of cyber crime

- In today's techno-savvy environment, the world is becoming more and more digitally sophisticated and so are the crimes.
- As the number of internet users is on the rise, the need for cyber laws and their application has also gathered great momentum.
- Cyber law is a framework created to give legal recognition to all risks arising out of the usage of computers and computer networks.
- Cyber law encompasses laws relating to:
 1. Cyber crimes
 2. Electronic and digital signatures
 3. Intellectual property
 4. Data protection and privacy

Legal perspective of cybercrime in India

In India, cybercrime is primarily governed by the Information Technology Act, 2000 (IT Act). This law was established to address various cyber offenses and provide a legal framework for electronic transactions, digital signatures, and data protection.

The purpose of the Indian IT Act(ITA) was to amend the Indian Penal Code(IPC).

Section Reference And Title	Chapter of the Act and Title	Crime	Punishment
Sec. 43 (Penalty for damage to computer system, etc.)	Chapter IX Penalties and Adjudication	Damage to computer system, etc.	Compensation for Rs. 1 Crore.
Sec. 66 (Hacking with computer system)	Chapter XI Offences	Hacking (With intent or knowledge).	Fine of Rs. 2 lakhs and imprisonment for 3 years.
Sec. 67 (Publishing of information which is obscene in electronics form).	Chapter XI Offences	Publication of obscene material in electronic form.	Fine of Rs. 1 lakh, of imprisonment for 5 years and double conviction on second offence.
Sec. 68 (Power of controller to give directions).	Chapter XI Offences	Not complying with directions of controller.	Fine up to Rs. 2 lakhs and imprisonment of 3 years.
Sec. 70(Protected system)	Chapter XI Offences	Attempting or securing access to computer of another person without his/her knowledge.	Imprisonment up to 10 years.
Sec.72(Penalty for breach of confidentiality and privacy)	Chapter XI Offences	Attempting or securing access to computer for breaking confidentiality of the information of computer.	Fine up to Rs. 1 lakh and imprisonment up to 2 years.
Sec.73(Penalty for publishing digital signature Certificates false in certain particulars)	Chapter XI Offences	Publishing false digital signatures, false in certain particulars.	Fine of Rs. 1 lakh or imprisonment of 2 years or both.
Sec.74(Publication for fraudulent purpose)	Chapter XI Offences	Publication of Digital Signatures for fraudulent purpose.	Imprisonment for the term 2 years and fine of Rs. 1 lakh.

The key provisions under the Indian ITA 2000

Amendments and Updates

- The IT Act has undergone amendments over the years to address emerging cyber threats and strengthen cybercrime provisions.
- For example, the Information Technology (Amendment) Act, 2008 introduced additional provisions to tackle cyber terrorism, data privacy, and intermediary liability.
- It is important to consult with legal professionals or refer to official sources for comprehensive and up-to-date information on the legal aspects of cybercrime in India.